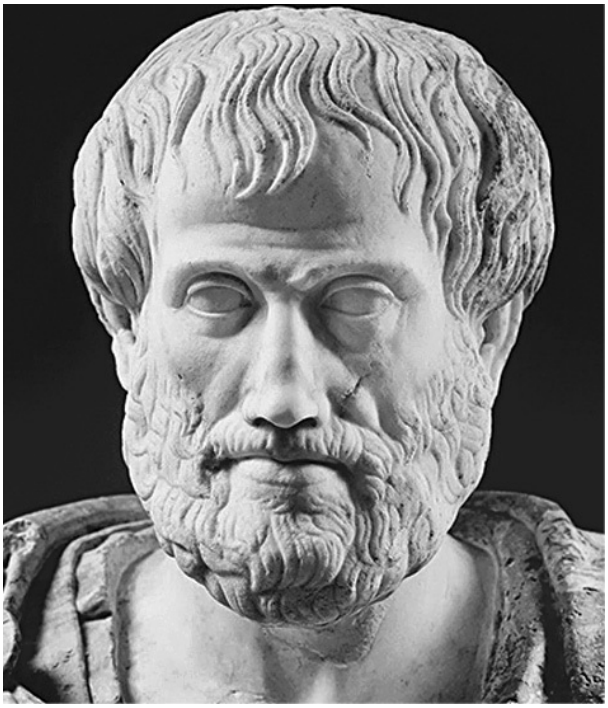
time must exist: “true” time that passes *regardless*, independently of things and of their changes. If all things remained motionless and even the movements of our souls were to be frozen, this time would continue to pass, according to Newton, unaffected and equal to itself: “true” time. It’s the exact opposite of what Aristotle writes.

“True” time, says Newton, is not directly accessible—only indirectly, through calculation. It is not the same as that given by days, because “the natural days are truly unequal, though they are commonly consider’d as equal, and used for a measure of time: Astronomers correct this inequality that they may measure the celestial motions by a more accurate time.”[45]

So, who is right: Aristotle or Newton? Two of the most acute and profound investigators of nature that the world has ever seen are proposing two opposite ways of thinking about time. Two giants are pulling us in opposite directions.[46]

Time is only a way of measuring how things change, as Aristotle would have it—or should we be thinking that an absolute time exists that flows by itself, independently of things? The question we should really be asking is this: which of these two ways of thinking about time helps us to understand the world better? Which of the two conceptual schemes is more efficient?



Aristotle: Time is nothing other than the measurement of change.

For a few centuries, reason seemed to come down on the side of Newton. Newton’s model, based on the idea of a time independent of things, has